



(Academic Session: 2024 - 2025)

ONLINE MAJOR TEST SERIES CONTACT PROGRAMME

Course : NEET OLTS
(ENTHUSE, LEADER/ACHIEVER)

PHASE-1
Batch: OLTS

Test Pattern
(NEET)

Date: 24-11-2024

Paper Type: Semi Major Test-1

READ THE INSTRUCTIONS CAREFULLY

Important Instructions:

1. The test is of 3 Hour 20 minutes duration and it contains 200 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
2. In this Test, subject Physics will consist of two sections (Section A & Section B). Section A will consist of 35 questions (all questions are mandatory) and Section B will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 3 attempted questions will be considered for marking.
3. Subject Chemistry will consist of two sections (Section A & Section B). Section A will consist of 35 questions (all questions are mandatory) and Section B will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 3 attempted questions will be considered for marking.
4. Subject Biology-I & Biology-II will consist of two sections (Section A & Section B). Section A will consist of 35 questions (all questions are mandatory) and Section B will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

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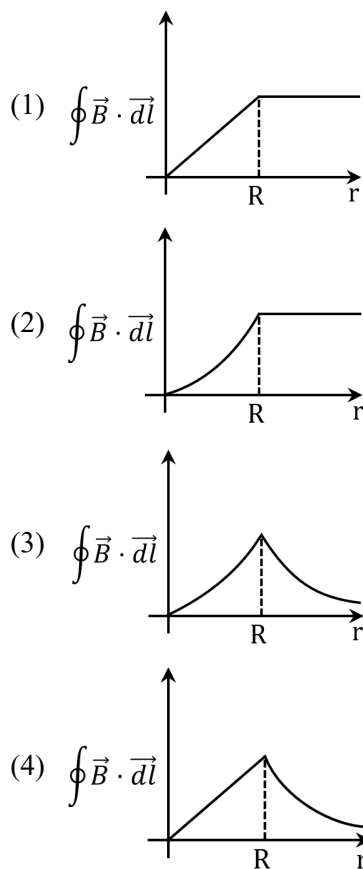


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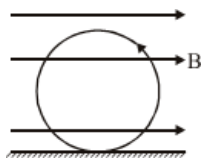
SECTION-A

Attempt All 35 questions

1. A cylindrical wire of radius R is carrying current i uniformly distributed over its cross-section. If a circular loop of radius r is taken as amperian loop, then the variation value of $\oint \vec{B} \cdot d\vec{l}$ over this loop with radius r of loop will be best represented by



2. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current $i = 4 \text{ A}$. A horizontal magnetic field $B = 10 \text{ T}$ is switched on at time $t = 0$ as shown in figure. The initial angular acceleration of ring is :-

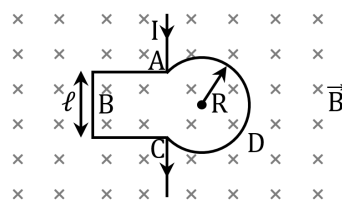


- (1) $40\pi \text{ rad/sec}^2$ (2) $20\pi \text{ rad/sec}^2$
 (3) $5\pi \text{ rad/sec}^2$ (4) $15\pi \text{ rad/sec}^2$

3. Which of the following statements is incorrect about hysteresis?

- (1) This effect is common to all ferromagnetic substances.
 (2) The hysteresis loop area is proportional to the thermal energy developed per unit volume of the material.
 (3) The hysteresis loop area is independent of the thermal energy developed per unit volume of the material.
 (4) The shape of the hysteresis loop is characteristic of the material.

4. The figure shows a conducting loop ABCDA placed in a uniform magnetic field (strength B) perpendicular to its plane. The part ABC is the three-fourth portion of the square of side length ℓ . The part ADC is a circular arc of radius R . The points A and C are connected to a battery which supply a current I to the circuit. The magnetic force on the loop due to the field B is :-

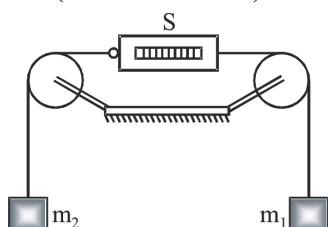


- (1) zero (2) $BI\ell$
 (3) $2BIR$ (4) $\frac{BI\ell R}{1+R}$

5. A thin circular ring of area A is perpendicular to uniform magnetic field of induction B . A small cut is made in the ring and a galvanometer is connected across the ends such that the total resistance of circuit is R . When the ring is suddenly squeezed to zero area, the charge flowing through galvanometer is :-

- (1) $\frac{BR}{A}$ (2) $\frac{AB}{R}$
 (3) ABR (4) $\frac{B^2 A}{R^2}$

6. An e.m.f. of 12 volts is induced in a given coil when the current in it changes at the rate of 48 amperes per minute. The self-inductance of the coil is :-
- (1) 0.25 henry (2) 15 henry
(3) 1.5 henry (4) 9.6 henry
7. The initial velocity of a particle is u and the acceleration is given by (kt) , where k is a positive constant. The distance travelled in time t is :
- (1) $s = ut^2 + kt^2$
(2) $s = ut + (kt^3/6)$
(3) $s = ut + (kt^3/2)$
(4) $s = (ut^2/2) + (kt^3/6)$
8. A rocket of mass 1000 kg exhausts gases at a rate of 4 kg/sec with velocity 3000 m/s. The thrust developed on the rocket is :-
- (1) 12000 N (2) 120 N
(3) 800 N (4) 200 N
9. A cricket ball of mass 250 g collides with a bat with velocity 10 m/s and returns with the same speed within 0.01 second. The force acted on bat is :-
- (1) 25 N (2) 50 N
(3) 250 N (4) 500 N
10. In the arrangement shown, the pulleys are fixed and ideal, the strings are light, $m_1 > m_2$ and S is a spring balance which is itself massless. The reading of S (in units of mass) is -



- (1) $m_1 - m_2$ (2) $\frac{1}{2} (m_1 + m_2)$
(3) $\frac{m_1 m_2}{m_1 + m_2}$ (4) $\frac{2m_1 m_2}{m_1 + m_2}$

11. The ratio of angular speed of hours hand and seconds hand of a clock is :-
- (1) 1 : 1 (2) 1 : 60
(3) 1 : 720 (4) 3600 : 1
12. If the radius of the earth is suddenly contracts to half of its present value, then the duration of day will be of
- (1) 6 hours (2) 12 hours
(3) 18 hours (4) 24 hours
13. While measuring diameter of wire using screw gauge the following readings were noted. Main scale reading is 1mm and circular scale reading is equal to 42 divisions. Pitch of screw gauge is 1mm and it has 100 divisions on circular Scale. The diameter of the wire is $\frac{x}{50}$ mm. The value of x is -
- (1) 142 (2) 71
(3) 42 (4) 21
14. A body crosses the topmost point of a vertical circle with a critical speed. Its centripetal acceleration, when the string is horizontal will be:-
- (1) $6g$ (2) $3g$
(3) $\sqrt{10}g$ (4) g
15. Two identical 5 kg blocks are moving with same speed of 2 m/s towards each other along a frictionless horizontal surface. The two blocks collide, stick together and come to rest. Consider the two blocks as a system. The work done by external and internal forces are respectively :-
- (1) 0, 0 (2) 0, 20 J
(3) 0, -20 J (4) 20 J, -20 J
16. The apparent weight of a person inside a lift is w_1 when lift moves up with a certain acceleration and is w_2 when lift moves down with same acceleration. The weight of the person when lift moves up with constant speed is :

- (1) $\frac{w_1 + w_2}{2}$ (2) $\frac{w_1 - w_2}{2}$
(3) $2w_1$ (4) $2w_2$

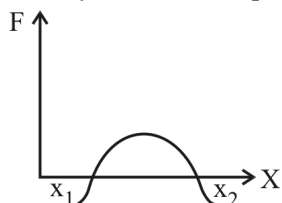
17. A ball A is thrown vertically upwards with speed u . At the same instant another ball B is dropped from height h . At times t , the speed of A relative to B is :-

(1) u (2) $u - 2gt$
 (3) $\sqrt{u^2 - 2gh}$ (4) $u - gt$

18. A body falls from a height of 16 m and rebounds to a height of 4 m. The coefficient of restitution is :

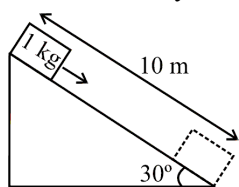
(1) $\frac{1}{4}$ (2) $\frac{1}{2}$
 (3) $\frac{3}{4}$ (4) $\frac{1}{8}$

19. The force acting on a body moving along x-axis varies with the position of the particle as shown in the figure. The body is in stable equilibrium at -



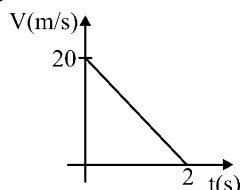
(1) $x = x_1$ (2) $x = x_2$
 (3) Both x_1 and x_2 (4) Neither x_1 nor x_2

20. Find work done by friction if block reaches to the end with constant velocity:-



(1) -50 J (2) $-50\sqrt{3} \text{ J}$
 (3) -100 J (4) None of these

21. Velocity - time graph of a particle of mass 2 kg moving in a straight line is as shown in figure. Work done by all the force on the particle is :-



(1) 400 J (2) -400 J
 (3) -200 J (4) 200 J

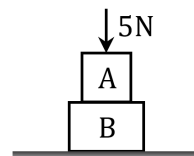
22. A solid cylinder is rolling down on an inclined plane of inclination 60° . What is its acceleration?

(1) $g/3$
 (2) $g/\sqrt{3}$
 (3) $\sqrt{\frac{2g}{3}}$
 (4) None of these

23. A constant torque of 1000 N-m turns a wheel of moment of inertia 200 kg-m^2 about an axis through its centre. Its angular velocity after 3 sec is :-

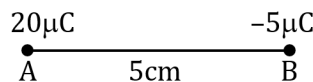
(1) 1 rad/sec (2) 5 rad/sec
 (3) 10 rad/sec (4) 15 rad/sec

24. If masses of blocks A and B are 3 kg and 8 kg respectively, then normal reaction between A and B :-



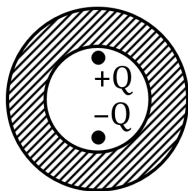
(1) 20N (2) 30N
 (3) 35N (4) 115N

25. Two particles A and B having charges $20 \mu\text{C}$ and $-5 \mu\text{C}$ respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?



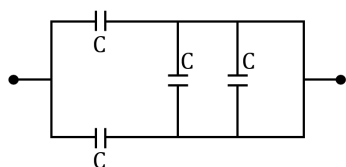
(1) At 5 cm from $20 \mu\text{C}$ on the left side of system
 (2) At 5 cm from $-5 \mu\text{C}$ on the right side
 (3) At 1.25 cm from $-5 \mu\text{C}$ between two charges
 (4) At midpoint between two charges

26. Shown in the figure are two point charges $+Q$ and $-Q$ inside the cavity of a spherical shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ_1 is the surface charge on the inner surface and Q_1 net charge on it and σ_2 the surface charge on the outer surface and Q_2 net charge on it then : -



- (1) $\sigma_1 \neq 0, Q_1 = 0$ (2) $\sigma_1 \neq 0, Q_1 = 0$
 $\sigma_2 = 0, Q_2 = 0$ $\sigma_2 \neq 0, Q_2 = 0$
- (3) $\sigma_1 = 0, Q_1 = 0$ (4) $\sigma_1 \neq 0, Q_1 \neq 0$
 $\sigma_2 = 0, Q_2 = 0$ $\sigma_2 \neq 0, Q_2 \neq 0$
27. Consider the force F on a charge ' q ' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F , if ' q ' is placed at distance r from the centre of the shell?

- (1) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r < R$
- (2) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2} > F > 0$ for $r < R$
- (3) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$ for $r > R$
- (4) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for all r
28. The equivalent capacitance of the combination shown is : -



- (1) $\frac{C}{2}$ (2) $4C$
- (3) $2C$ (4) $\frac{5}{3}C$

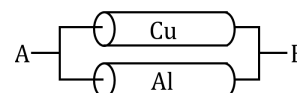
29. **Statement 1** : It is not possible to make a sphere of capacity 1 farad using a conducting material.

Statement 2 : It is possible for earth as its radius is 6.4×10^6 m.

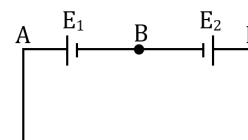
- (1) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
- (2) Statement 1 is false, Statement 2 is true.
- (3) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
- (4) Statement 1 is true, Statement 2 is false.
30. The electric current through a wire varies with time as $I = I_0 + \beta t$, where $I_0 = 20$ A and $\beta = 3$ A/s. The amount of electric charge crossed through a section of the wire in 20s is : -

- (1) 1600 C (2) 800 C
- (3) 80 C (4) 1000 C

31. A Copper (Cu) rod of length 25 cm and cross-sectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B. (Take Resistivity of Copper = $1.7 \times 10^{-8} \Omega\text{m}$ Resistivity of Aluminium = $2.6 \times 10^{-8} \Omega\text{m}$)



- (1) 2.170 mΩ (2) 1.420 mΩ
- (3) 0.0858 mΩ (4) 0.858 mΩ
32. Two cells are connected in opposition as shown. Cell E_1 is of 8V emf and 2Ω internal resistance; the cell E_2 is of 2V emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is _____ V.



- (1) 6V (2) -8V
- (3) -6V (4) 8V

33. Given below are two statements : One is labelled as

Assertion A and the other is labelled as **Reason R**.

Assertion A : Alloys such as constantan and manganin are used in making standard resistance coils.

Reason R : Constantan and manganin has very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both A and R are true and R is the correct explanation of A.
 (2) Both A and R are true but R is NOT the correct explanation of A.
 (3) A is true but R is false.
 (4) A is false but R is true.

34. The values of kinetic energy (K) and potential energy (U) are measured as follows

$$K = (100.0 \pm 2.0) \text{ J and } U = (200.0 \pm 1.0) \text{ J}$$

then the percentage error in the measurement of mechanical energy is :

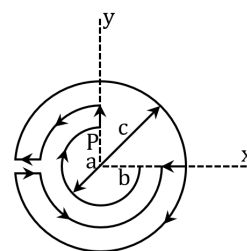
- (1) 2.5% (2) 0.5%
 (3) 1.0% (4) 1.5%
35. Least count of a vernier caliper is $\frac{1}{20N}$ cm. The value of one division on main scale is 1mm. Then the number of divisions of main scale that coincide with N divisions of vernier scale is -

- (1) $\left(\frac{2N-1}{20N}\right)$ (2) $\left(\frac{2N-1}{2}\right)$
 (3) $(2N-1)$ (4) $\left(\frac{2N-1}{2N}\right)$

SECTION-B

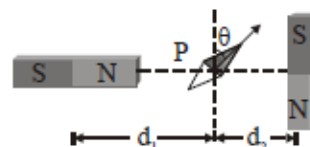
This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

36. For $c = 2a$, and $a < b < c$, the magnetic field at point P will be zero when



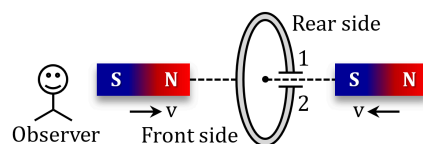
- (1) $a = b$ (2) $a = \frac{3}{5}b$
 (3) $a = \frac{5}{3}b$ (4) $a = \frac{1}{3}b$

37. Two magnets A and B are identical and these are arranged as shown in the figure. Their length is negligible in comparison to the separation between them. A magnetic needle is placed between the magnets at point P which gets deflected through an angle θ under the influence of magnets. The ratio of distance d_1 and d_2 will be. :-



- (1) $(2 \tan \theta)^{1/3}$ (2) $(2 \tan \theta)^{-1/3}$
 (3) $(2 \cot \theta)^{1/3}$ (4) $(2 \cot \theta)^{-1/3}$

38. The north and south poles of two identical magnets approach a coil containing a condenser, with equal speeds from opposite sides. Then :-

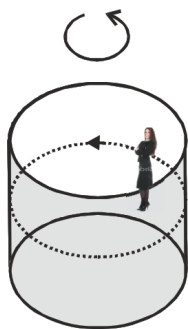


- (1) Plate 1 will be negative and plate 2 positive
 (2) Plate 1 will be positive and plate 2 negative
 (3) Both the plates will be positive
 (4) Both the plates will be negative

39. If P represents radiation pressure, c represents speed of light and Q represents radiation energy striking a unit area per second, then non-zero integers x , y and z such that $P^x Q^y c^z$ is dimensionless, can be :-

- (1) $x = 1, y = 1, z = -1$
- (2) $x = 1, y = -1, z = 1$
- (3) $x = -1, y = 1, z = 1$
- (4) $x = 1, y = 1, z = 1$

40. A girl finds herself stuck with her back to the wall of a cylinder rotating about its axis. Which diagram correctly shows the forces acting on her?



- (1)
- (2)
- (3)
- (4)

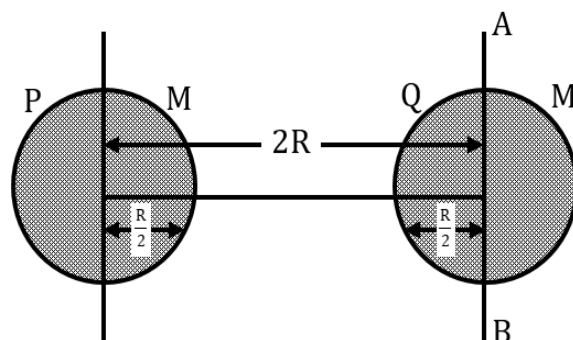
41. Water falling from a 50 m high fall is to be used for generating electric energy. If 1.8×10^5 kg of water falls per hour and half the gravitational potential energy can be converted into electric energy, how many 100 W lamps can be lit :-

- (1) 75
- (2) 100
- (3) 125
- (4) 150

42. A rod of length 3m and its mass per unit length is directly proportional to the distance x from its one end. The center of gravity of the rod from that end will be at

- (1) 1.5 m
- (2) 2 m
- (3) 2.5 m
- (4) 3.0 m

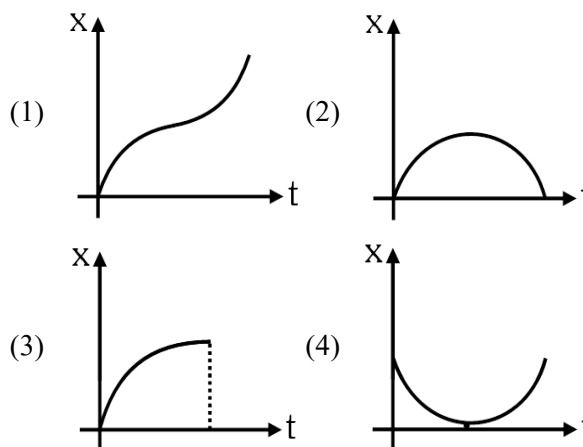
43. Two spheres each of mass M and radius $R/2$ are connected with massless rod of length $2R$ as shown in the figure.



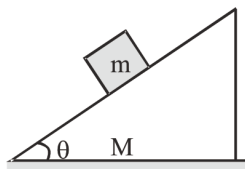
The moment of inertia of the system about an axis passing through the centre of one of the spheres and perpendicular to the rod is

- (1) $\frac{21}{5} MR^2$
- (2) $\frac{2}{5} MR^2$
- (3) $\frac{5}{2} MR^2$
- (4) $\frac{5}{21} MR^2$

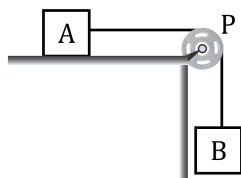
44. For a ball thrown vertically upwards and returning back to ground, which will be its distance-time graph.



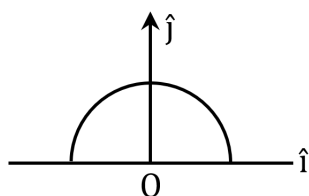
45. In Fig. shown, the block of mass m is at rest relative to the wedge of mass M and the wedge is at rest with respect to ground. This implies that :-



- (1) Net force applied by m on M is mg .
 - (2) Normal force applied by m on M is mg .
 - (3) Force of friction applied by m on M is mg .
 - (4) None of above
46. The blocks A and B are arranged as shown in the figure. The pulley is frictionless. The mass of A is 10 kg. The coefficient of friction between block A and horizontal surface is 0.20. The minimum mass of B to start the motion will be :-

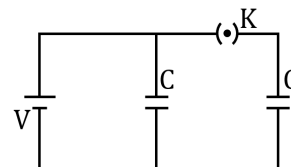


- (1) 2 kg
 - (2) 0.2 kg
 - (3) 5 kg
 - (4) 10 kg
47. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is :-

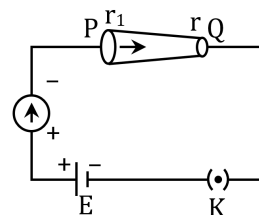


- (1) $\frac{q}{4\pi^2 \epsilon_0 r^2} \hat{j}$
- (2) $-\frac{q}{4\pi^2 \epsilon_0 r^2} \hat{j}$
- (3) $-\frac{q}{2\pi^2 \epsilon_0 r^2} \hat{j}$
- (4) $\frac{q}{2\pi^2 \epsilon_0 r^2} \hat{j}$

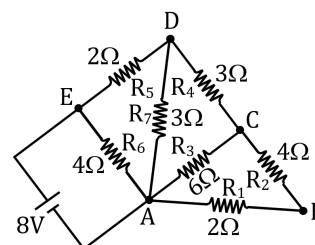
48. A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1 . Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2 . The ratio E_1/E_2 will be :-



- (1) $\frac{1}{10}$
 - (2) $\frac{2}{5}$
 - (3) $\frac{5}{13}$
 - (4) $\frac{5}{26}$
49. In the given figure, a battery of emf E is connected across a conductor PQ of length ' ℓ ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$). Choose the correct option as one moves from P to Q :-



- (1) Drift velocity of electron increases.
 - (2) Electric field decreases.
 - (3) Electron current decreases.
 - (4) All of these.
50. The current flowing through R_2 is :-



- (1) $\frac{2}{3}$ A
- (2) $\frac{1}{4}$ A
- (3) $\frac{1}{2}$ A
- (4) $\frac{1}{3}$ A

SECTION-A

Attempt All 35 questions

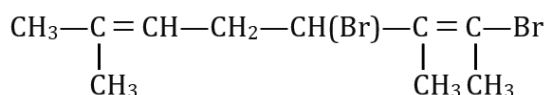
51. Which of the following nitrogen containing compound does not give Lassaing's test ?

- (1) Phenyl hydrazine (2) Glycine
(3) Urea (4) Hydrazine

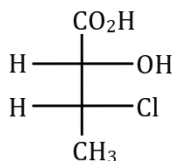
52. The adsorbent used in adsorption chromatography is/are

- (1) silica gel (2) quick lime
(3) magnesia (4) None of above

53. Total No. of stereoisomer possible for the given structure :



- (1) 2 (2) 4
(3) 8 (4) 6

54. The absolute configuration of  is :

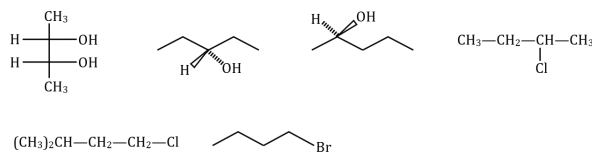
- (1) (2S, 3S) (2) (2R, 3R)
(3) (2R, 3S) (4) (2S, 3R)

55. **Assertion [A]** : $\text{S}_{\text{N}}2$ reaction of $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$ occurs more readily than the $\text{S}_{\text{N}}2$ reaction of $\text{CH}_3\text{CH}_2\text{Br}$.

Reason [R] : The partially bonded unhybridized p-orbital that develops in the trigonal bipyramidal transition state is stabilized by conjugation with the phenyl ring.

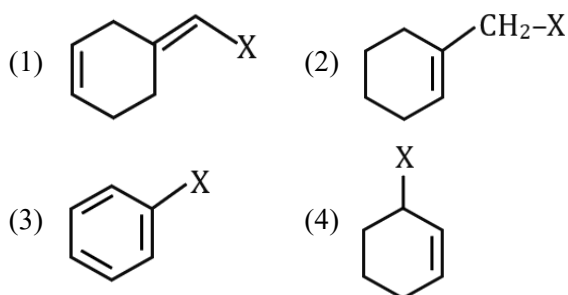
- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

56. Total number of optically active compounds from the following is :-

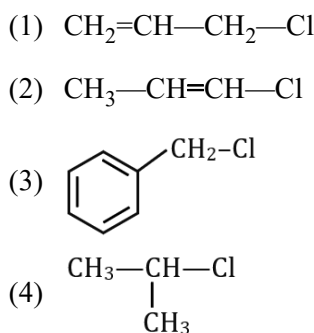


- (1) 4 (2) 3
(3) 2 (4) 6

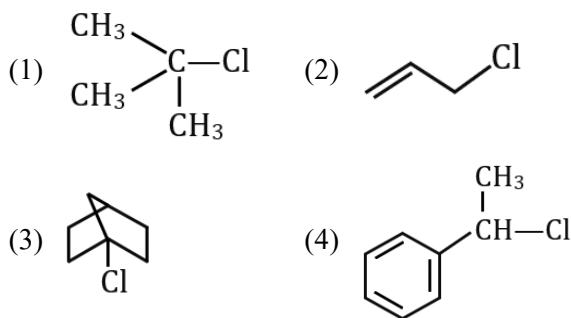
57. Example of Vinylic halide is :



58. Which among the following halide will not show $\text{S}_{\text{N}}1$ reaction



59. Which of the following compound is inactive towards $\text{S}_{\text{N}}1$ reaction



69. Which of the following configurations of ions have zero CFSE in both strong and weak field ligands in octahedral complex ?
 (1) d^8 (2) d^4
 (3) d^6 (4) d^{10}
70. Facial-meridional isomerism is associated with which one of the following complexes ?
 (1) $[\text{Pt}(\text{py})_2(\text{NH}_3)_2\text{Cl}]\text{SO}_4$
 (2) $[\text{Cr}(\text{gly})_3]$
 (3) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (4) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
71. Ethylene diaminetetraacetate (EDTA) ion is :
 (1) Hexadentate ligand with four "O" and two "N" donor atoms
 (2) Unidentate ligand
 (3) Bidentate ligand with two "N" donor atoms
 (4) Tridentate ligand with three "N" donor atoms
72. Which of the following is not an ionic carbide :-
 (1) CaC_2 (2) Al_4C_3
 (3) SiC (4) Be_2C
73. Order of relative stability of +1 oxidation state in Boron family :-
 (1) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (2) $\text{Al} > \text{Ga} > \text{In} < \text{Tl}$
 (3) $\text{Al} = \text{Ga} = \text{In} = \text{Tl}$ (4) $\text{Al} < \text{Ga} > \text{In} < \text{Tl}$
74. Which of the following compound does not show Hydrolysis :-
 (1) NF_3 (2) CCl_4
 (3) SF_6 (4) All of these
75. Which of the following oxy acids of sulphur contains a sulphur-sulphur single bond ?
 (1) $\text{H}_2\text{S}_2\text{O}_7$ (2) $\text{H}_2\text{S}_2\text{O}_8$
 (3) $\text{H}_2\text{S}_2\text{O}_5$ (4) H_2SO_5
76. In Arrhenius equation $k = Ae^{-E_a/RT}$, as $\lim_{T \rightarrow \infty} \log_{10} K$ equals to :
 (1) $\ln A$ (2) A
 (3) $\log_{10} A$ (4) none of these
77. **Statement-1:** Time taken for the completion of 75% of a I^{st} order reaction is double that its $t_{1/2}$.
Statement-2: Time taken for completion of any fraction of a I^{st} order reaction is a constant value.
 (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (3) Statement-1 is True, Statement-2 is False.
 (4) Statement-1 is False, Statement-2 is True.
78. **Statement-1:** A catalyst provides an alternative path to the reaction in which conversion of reactants into products take place quickly.
Statement-2: The catalyst forms an activated complex of lower potential energy, with the reactants by which more number of molecules are able to cross the barrier per unit of time.
 (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (3) Statement-1 is True, Statement-2 is False.
 (4) Statement-1 is False, Statement-2 is True.
79. Correct statements about a voltaic (galvanic) cell include which of the following?
 P. Oxidation occurs at the anode.
 Q. Electrons flow from the cathode to the anode.
 (1) P only (2) Q only
 (3) Both P and Q (4) Neither P nor Q

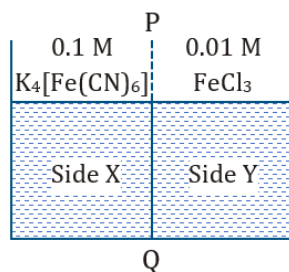
80. A gas Cl_2 at 1 atm is bubbled through a solution containing a mixture of 1 M Br^- and 1 M F^- at 25°C . If the reduction potential order is $\text{F} > \text{Cl} > \text{Br}$, then :

- (1) Cl_2 will oxidise Br^- and not F^-
- (2) Cl_2 will oxidise F^- and not Br^-
- (3) Cl_2 will oxidise both Br^- and F^-
- (4) Cl_2 will reduce both Br_2 and F_2

81. There is AB_3 and sucrose solution with 0.1 M concentration each. If osmotic pressure of AB_3 and sucrose solution is 0.72 atm and 0.24 atm respectively at same temperature. The fraction of AB_3 dissociated will be :

- (1) 0.25
- (2) 0.50
- (3) 0.67
- (4) 0.90

82. FeCl_3 on reaction with $\text{K}_4[\text{Fe}(\text{CN})_6]$ in aq. solution gives blue colour. These are separated by a semipermeable membrane PQ as shown. Due to osmosis there is :



- (1) blue colour formation in side X
- (2) blue colour formation in side Y
- (3) blue colour formation in both of the sides X and Y
- (4) no blue colour formation

83. Moles of K_2SO_4 to be dissolved in 12 mol water to lower its vapour pressure by 10 mm Hg at a temperature at which vapour pressure of pure water is 50 mm is : (Assume complete dissociation)

- (1) 3 mol
- (2) 2 mol
- (3) 1 mol
- (4) 0.5 mol

84. The index for the gold number is

- (1) Protective power of lyophilic colloid
- (2) Purity of gold
- (3) Metallic gold
- (4) Electroplated gold

85. Phenolphthalein color in basic medium is

- (1) Pink
- (2) Orange
- (3) Yellow
- (4) Colourless

SECTION-B

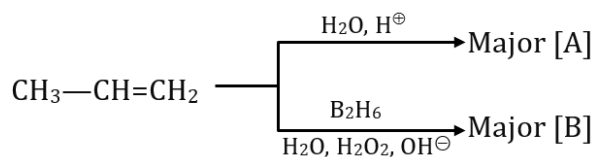
This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

86. Match the following :

Column-I (Test/Method)		Column-II (Reagent)	
(I)	Lucas Test	(A)	$\text{C}_6\text{H}_5\text{SO}_2\text{Cl/aq. KOH}$
(II)	Dumas method	(B)	$\text{HNO}_3/\text{AgNO}_3$
(III)	Kjeldahl's method	(C)	CuO/CO_2
(IV)	Hinsberg test	(D)	Conc. HCl and ZnCl_2
		(E)	H_2SO_4

- (1) (I)-(D), (II)-(C), (III)-(B), (IV)-(A)
- (2) (I)-(B), (II)-(D), (III)-(E), (IV)-(A)
- (3) (I)-(D), (II)-(C), (III)-(E), (IV)-(A)
- (4) (I)-(B), (II)-(A), (III)-(C), (IV)-(D)

87. Identify the product A and product B in the following set of reactions.



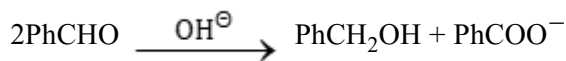
- (1) A = $\text{CH}_3-\text{CH}_2-\text{CH}_3$, B = $\text{CH}_3-\text{CH}_2-\text{CH}_3$
 (2) A = $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$, B = $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}_2}$
 (3) A = $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$, B = $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$
 (4) A = $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$, B = $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$

88. **Assertion (A) :-** Alcohols act as both nucleophiles and electrophiles.

Reason (R) :- Alcohols react with active metal sodium, potassium and aluminium to form alkoxides and Hydrogen gas.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 (3) (A) is correct but (R) is not correct.
 (4) (A) is not correct but (R) is correct.

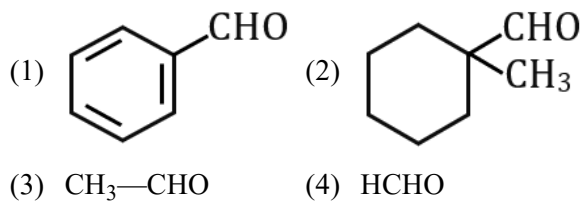
89. In Cannizzaro reaction given below :



the slowest step is :

- (1) The transfer of hydride to the carbonyl group
 (2) The removal of proton from the carboxylic group
 (3) The deprotonation of PhCH_2OH
 (4) The attack of OH^- at the carboxylic group.

90. Cannizzaro's reaction is not given by :



91. Given below are two statements :

Statement (I) : The NH_2 group in Aniline is ortho and para directing and a powerful activating group.

Statement (II) : Aniline does not undergo Friedel-Craft's reaction (alkylation and acylation).

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both Statement I and Statement II are correct
 (2) Statement I is correct but Statement II is incorrect
 (3) Statement I is incorrect but Statement II is correct
 (4) Both Statement I and Statement II are incorrect

92. KMnO_4 acts as an oxidizing agent in :

- (1) Acidic medium only
 (2) Neutral and acidic media
 (3) Neutral and alkaline media
 (4) Neutral, acidic, and alkaline media

93. **Statement-I :-** Geometrical isomers of the complex $[\text{M}(\text{NH}_3)_4\text{Br}_2]$ are optically inactive.

Statement-II : Both geometrical isomers of $[\text{M}(\text{NH}_3)_4\text{Br}_2]$ have plane of symmetry.

- (1) Both Statements are true and statement-2 is the correct explanation for statement-1
 (2) Both Statements are true but statement-2 is not the correct explanation for statement-1
 (3) Statement-1 is true, statement-2 is false
 (4) Statement-1 is false, statement-2 is true

94. Which of the following complexes has magnetic moment of 2.83 BM ?

- (1) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
- (2) $[\text{CoCl}_6]^{3-}$
- (3) $[\text{Ni}(\text{CN})_4]^{2-}$
- (4) $[\text{Cr}(\text{NH}_3)_6]^{3+}$

95. 3 centered 2 electron bond is present in :-

- (1) Be_2H_4
- (2) $\text{Al}_2(\text{CH}_3)_6$
- (3) B_2H_6
- (4) All of these

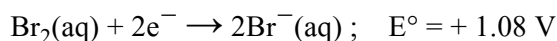
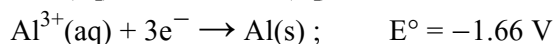
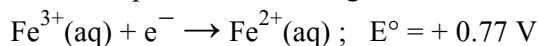
96. For the reaction; $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$, the rate law is $\text{Rate} = k[\text{H}_2][\text{I}_2]$.

Which of the mechanisms given is consistent with this rate law?

I	II
$\text{I}_2 \rightleftharpoons \text{I} + \text{I}$ (fast) $\text{H}_2 + \text{I} + \text{I} \rightarrow 2\text{HI}$ (slow)	$\text{I}_2 \rightleftharpoons \text{I} + \text{I}$ (slow) $\text{H}_2 + \text{I} + \text{I} \rightarrow 2\text{HI}$ (fast)

- (1) I only
- (2) II only
- (3) Either I or II
- (4) Neither I nor II

97. Electrode potential data are given below:



Based on the data given above reducing power of Fe^{2+} , Al and Br^- will increase in the order :

- (1) $\text{Br}^- < \text{Fe}^{2+} < \text{Al}$
- (2) $\text{Fe}^{2+} < \text{Al} < \text{Br}^-$
- (3) $\text{Al} < \text{Br}^- < \text{Fe}^{2+}$
- (4) $\text{Al} < \text{Fe}^{2+} < \text{Br}^-$

98. Given that,

$$E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44 \text{ V}; \quad E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.77 \text{ V}$$

If Fe^{3+} and Fe solid are kept together then :

- (1) Fe^{3+} increases
- (2) Fe^{3+} decreases
- (3) $\text{Fe}^{2+}/\text{Fe}^{3+}$ remains unchanged
- (4) Fe^{2+} decreases

99. For 0.1 m aq. solution of given electrolytes correct order of their increasing order of boiling point :

- (1) $\text{CH}_3\text{COOH} < \text{HCOOH} < \text{NaCl} < \text{Na}_2\text{SO}_4$
- (2) $\text{HCl} < \text{HCOOH} < \text{NaCl} < \text{Na}_2\text{SO}_4$
- (3) $\text{HCOOH} < \text{CH}_3\text{COOH} < \text{NaCl} < \text{Na}_2\text{SO}_4$
- (4) $\text{Na}_2\text{SO}_4 < \text{NaCl} < \text{HCOOH} < \text{CH}_3\text{COOH}$

100. 10 ml of 10 M H_2SO_4 is mixed to 100 ml 1M NaOH solution. The resultant solution will be

- (1) Acidic
- (2) Neutral
- (3) Weakly alkaline
- (4) Strongly alkaline

SECTION-A

Attempt All 35 questions

101. **Statement-I** : All living organisms possess metabolic reactions.

Statement-II : Each name have two components generic name and the specific epithet.

- (1) Both Statement-I and Statement-II are true.
- (2) Both Statement-I and Statement-II are false.
- (3) Statement-I is true, but Statement-II is false.
- (4) Statement-I is false, but Statement-II is true.

102. Who is known as the "Father of Taxonomy"?

- (1) Aristotle
- (2) Linnaeus
- (3) Darwin
- (4) Lamarck

103. Identify the incorrect statement(s) :-

- (I) Growth is a defining feature of living organisms.
- (II) All living organisms show cellular organization.
- (III) Reproduction is a defining characteristic of all living organisms.
- (IV) All living organisms exhibit metabolism.

- (1) I and II
- (2) III only
- (3) II and IV
- (4) I, II, and III

104. In which type of phyllotaxy each node bear two leaves?

- (1) Alternate
- (2) Opposite
- (3) Whorled
- (4) Spiral

105. Which part of a flower develops into fruit?

- (1) Ovule
- (2) Ovary
- (3) Petal
- (4) Anther

106. Identify the incorrect statement(s) :-

- (1) In compound leaves, the leaf blade is divided into leaflets.
- (2) The outer covering of endosperm separates the embryo by a proteinous layer called aleurone layer.
- (3) Ovary is one-chambered but it becomes two chambered due to the formation of the false septum in tomato.
- (4) The style connects the ovary to the stigma.

107. The type of inflorescence wherein the main axis has limited growth and ends in a flower is termed :-

- (1) Racemose
- (2) Cymose
- (3) Hypanthodium
- (4) Cyathium

108. In a monocot stem, which of the following structures is absent?

- (1) Ground tissue
- (2) Epidermis
- (3) Vascular bundles
- (4) Cambium

109. **Statement-I** : When xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii, the arrangement is called radial such as in stem.

Statement-II : The conjoint vascular bundles usually have the phloem located only on the outer side of xylem.

- (1) Both Statement-I and Statement-II are correct.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Both Statement-I and Statement-II are incorrect.
- (4) Statement-I is incorrect but Statement-II is correct.

110. Identify the correct statement :-

- (1) Initiation of lateral roots and vascular cambium during the secondary growth takes place in hypodermal cells.
- (2) The pith is small or inconspicuous in monocot root.
- (3) As compared to the monocot root, dicot root have fewer xylem bundles.
- (4) The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of conjunctive tissue.

111. The four elements called "Big-four" which make up 95% of all elements found in a living system are :-

- (1) C, H, O, N
- (2) C, H, O, P
- (3) C, H, O, S
- (4) C, N, O, P

112. Macromolecules are :-

- (1) Nucleic acids, proteins and polysaccharides
- (2) Nucleic acids and monosaccharides
- (3) Amino acids and polysaccharides
- (4) Amino acids, lipids and nucleotides

113. Fructose is a ketose sugar and also called :-

- (1) An aldose
- (2) Fruit sugar
- (3) Cane sugar
- (4) Corn sugar

114. Invert sugar is mixture of :-

- (1) Maltose and fructose
- (2) Glucose and galactose
- (3) Glucose and fructose
- (4) All of the above

115. Identify the incorrect statement about Z scheme :-

- (1) Z shaped is formed when all the carriers are placed in sequence on a redox potential scale.
- (2) PS I are excited when they receive red light of wavelength 700 nm.
- (3) PS II absorbs 680 nm of blue light causing electrons to become excited.
- (4) Addition of electron (downhill) reduces NADP^+ to $\text{NADPH} + \text{H}^+$

116. _____ is the synthesis of ATP from ADP and inorganic phosphate in the presence of light.

- (1) Phosphorylation
- (2) Photo phosphorylation
- (3) Light reaction
- (4) Dark reaction

117. Photophosphorylation occurs due to cyclic flows of electrons when only PS I is functional in :-

- (1) Stroma lamellae
- (2) Grana lamellae
- (3) Stroma
- (4) Lumen

118. Which of the following is a first product of the Calvin cycle?

- (1) Glucose
- (2) Sucrose
- (3) ATP
- (4) Glyceraldehyde-3-phosphate

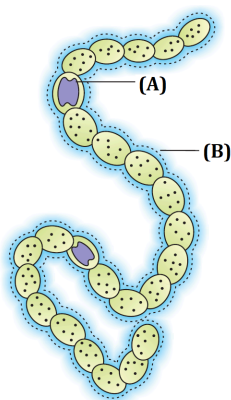
119. How many ATP are formed in ETS from reduced NAD generated in one turn cycle of krebs' cycle?

- (1) 3
- (2) 6
- (3) 12
- (4) 9

120. Chemiosmotic hypothesis of ATP synthesis in respiration based on :-

- (1) Proton gradient between lumen and stroma.
- (2) Proton gradient between matrix and inner membrane.
- (3) Proton gradient between matrix and inner membrane space.
- (4) Proton gradient between matrix and inter membrane space.

121. What is shown in the figure :-



- (1) A → Heterocyst, B → Mucilaginous sheath
- (2) A → Akinete, B → Filament
- (3) A → Akinete, B → Hormogonia
- (4) A → Akinete, B → Trichome

122. Match the column and find the correct answer :-

	Column-I		Column-II
(A)	<i>Streptomyces</i>	(i)	Food poisoning
(B)	<i>Rhizobium</i>	(ii)	Source of antibiotics
(C)	<i>Nitrosomonas</i>	(iii)	Nitrogen fixation
(D)	<i>Acetobacter</i>	(iv)	Nitrification
		(v)	Vinegar synthesis

- (1) A-iv, B-v, C-i, D-iii
- (2) A-v, B-i, C-iii, D-iv
- (3) A-ii, B-iii, C-iv, D-v
- (4) A-iv, B-iii, C-i, D-v

123. The unicellular eukaryotic organisms were placed in :-

- (1) Protista
- (2) Monera
- (3) Fungi
- (4) Animalia

124. Fungi differ from algae in having :-

- (1) Mainly cellulosic cell wall
- (2) Mainly cell wall of chitin
- (3) Unicellular nonjacketed sex organ
- (4) Starch as reserve food

125. Match the following :-

	Column-I		Column-II
(i)	Phaeophyceae	(A)	Fucoxanthin
(ii)	Cyanophyceae	(B)	Phycoerythrin
(iii)	Rhodophyceae	(C)	Chl a, b
(iv)	Diatoms	(D)	Chrysolaminarin

- (1) i-A, ii-B, iii-C, iv-D
- (2) i-A, ii-C, iii-B, iv-D
- (3) i-C, ii-B, iii-A, iv-D
- (4) i-B, ii-A, iii-D, iv-C

126. Characteristic of angiosperms which distinguish them from gymnosperms :-

- (1) Presence of fruits and flowers
- (2) Double fertilization and triploid endosperm formed after double fertilization
- (3) Companion cells in phloem and vessels in xylem
- (4) All of the above

127. *Selaginella* and *Salvinia* are considered to represent a significant step toward evolution of seed habit because :-

- (1) Female gametophyte is free and gets dispersed like seeds.
- (2) Female gametophyte lacks archegonia.
- (3) Megaspores possess endosperm and embryo surrounded by seed coat.
- (4) Embryo develops in female gametophyte which is retained on parent sporophyte.

128. Assertion : In Eukaryotic cells there is an extensive compartmentalisation of cytoplasm.

Reason : There is a presence of membrane bound organelles in eukaryotic cells.

- (1) Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (2) Assertion and Reason are correct and Reason is not the correct explanation of Assertion.
- (3) Assertion is correct and Reason is incorrect.
- (4) Assertion is incorrect and Reason is correct.

129. Which statement is not correct about integral and peripheral proteins :-

- (1) They are considered as a membrane protein.
- (2) They are classified based on ease of extraction.
- (3) Peripheral proteins lie on the surface of membrane while integral proteins are partially buried in the membrane.
- (4) Peripheral proteins lie on the surface while integral proteins are partially or totally buried in the membrane.

130. What is the cause of cell division?

- (1) Increase in surface area/volume ratio due to growth of cell.
- (2) Nucleo-cytoplasmic ratio decreases due to increase in size of cell.
- (3) Disturbance in lipid and protein ratio.
- (4) All of the above

131. In disjunction (separation) of sister chromatids at anaphase :-

- (1) Centromere moves first and arms trail behind.
- (2) Arms move first and centromere trails behind.
- (3) Any of the above cases.
- (4) Centromere does not move in anaphase.

132. The sequence of substages of prophase-I are :-

- (1) Leptotene — Zygotene — Pachytene — Diplotene — Diakinesis
- (2) Leptotene — Zygotene — Pachytene — Diakinesis — Diplotene
- (3) Leptotene — Zygotene — Pachytene — Diakinesis
- (4) Leptotene — Pachytene — Diplotene — Zygotene — Diakinesis

133. The phenomenon of apical dominance can be overcome by exogenous application of :-

- (1) ABA
- (2) Gibberellins
- (3) Cytokinins
- (4) Ethylene

134. The first natural cytokinin of plants is :-

- (1) Zeatin
- (2) Kinetin
- (3) Dihydroxyzeatin
- (4) Riboxylzeatin

135. Bakanae disease in Japan was due to a fungus known as :-

- (1) *Gibberella fujikori*
- (2) *Aspergillus flavus*
- (3) Both (1) and (2)
- (4) *Plasmopara viticola*

SECTION-B

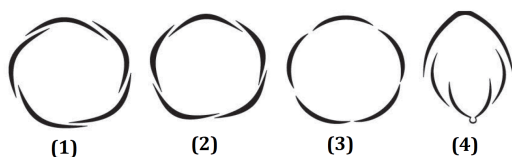
This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

136. **Assertion** : Nomenclature provides each organism a scientific name.

Reason : Scientific names are universally accepted.

- (1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
- (3) Assertion is true, but Reason is false.
- (4) Assertion is false, but Reason is true.

137. Out of four aestivations of petals given below, which one is found in Malvaceae :-



- (1) 1
- (2) 2
- (3) 3
- (4) 4

138. **Statement-I** : Sclerenchyma cells are dead at maturity.

Statement-II : Sclerenchyma cells have lignified cell walls that provide rigidity.

- (1) Both Statement-I and Statement-II are correct.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Both Statement-I and Statement-II are incorrect.
- (4) Only Statement-II is correct.

139. Amino acids usually exist in the form of Zwitter ions. This means that they consist of :-

- (1) The basic NH_2 group and acidic COOH group.
- (2) The basic NH_3^+ group and acidic COO^- group.
- (3) Basic COO^- group and acidic NH_3^+ group.
- (4) None of the above

140. Cellulose in plant cell wall is made up of :-

- (1) Unbranched chain of glucose molecules linked by α , $1 \rightarrow 6$ glycosidic bond.
- (2) Unbranched chain of glucose molecules linked by β , $1 \rightarrow 4$ glycosidic bond.
- (3) Branched chain of glucose molecules linked by α , $\rightarrow 6$ glycosidic bond in straight chain & $\beta \rightarrow 1,4$ at the site of branching.
- (4) Branched chains have $\alpha \rightarrow 1,4$ bond and $\beta \rightarrow 1,6$ glycosidic bonds both.

141. Which statement is incorrect with respect of splitting of water :-

- (1) Splitting of water is associated with PS II
- (2) Water splits into 2H^+ , $[\text{O}]$ and electrons
- (3) Oxygen is one of the net products of splitting of water
- (4) Splitting of water process occurs on the outer side of the membrane of thylakoid

142.
$$\text{Pyruvic acid} + \text{CoA} + \text{NAD}^+ \xrightarrow[\text{Pyruvate dehydrogenase}]{(\text{A})} \text{Acetyl CoA} + \text{CO}_2 + \text{NADH} + \text{H}^+$$

A is $\rightarrow$

- (1) Coenzyme A
- (2) Mg^{+2}
- (3) K^+
- (4) ATP

143. Match the following and select the correct combination from the options given below :-

	Column-I (Kingdom)		Column-II (Class)
(A)	Plantae	(1)	Archaeobacteria
(B)	Fungi	(2)	Euglenoids
(C)	Protista	(3)	Phycomycetes
(D)	Monera	(4)	Algae

- (1) A-4, B-3, C-2, D-1
 (2) A-1, B-2, C-3, D-4
 (3) A-3, B-4, C-2, D-1
 (4) A-2, B-3, C-4, D-1

144. Mycorrhizae are symbiotic association of fungi with roots of higher plants. these are useful for plants due to their following attribute.

- (1) Fixing atmospheric nitrogen.
 (2) Enhancing absorption of nutrients from soil.
 (3) Killing atmospheric nitrogen in nearby area.
 (4) Providing susceptibility against abiotic stress.

145. Select the mismatch :-

- (1) *Pinus* – Coralloid roots
 (2) *Cycas* – Unbranched stem
 (3) *Salvinia* – Heterosporous
 (4) *Equisetum* – Homosporous

146. Which one is wrongly matched?

- (1) Unicellular organism — *Chlorella*
 (2) Gemma cups — *Marchantia*
 (3) Biflagellate zoospores — Brown algae
 (4) Uniflagellate gametes — *Polysiphonia*

147. **Assertion** : The cell wall provides a strong structural support to prevent the bacterium from bursting or collapsing.

Reason : Cell envelope including cell wall act together as a single protective unit.

- (1) Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (2) Assertion and Reason are correct and Reason is not the correct explanation of Assertion.
 (3) Assertion is correct and Reason is incorrect.
 (4) Assertion is incorrect and Reason is correct.

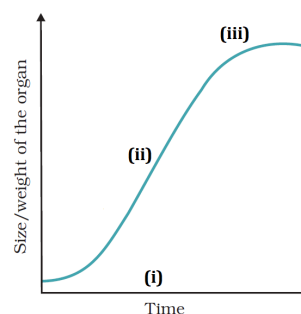
148. Which is not a function of endoplasmic reticulum :

- (1) Endoplasmic reticulum form endoskeleton of cell
 (2) Endoplasmic reticulum is the major site for synthesis of lipid.
 (3) Endoplasmic reticulum is concerned with detoxification of drugs, pollutants and steroids.
 (4) Endoplasmic reticulum perform the function of packaging material.

149. Bivalents are formed in :-

- (1) Diplotene (2) Pachytene
 (3) Zygotene (4) Interkinesis

150. Go through the following figure :-



Choose the correct labelling.

	(i)	(ii)	(iii)
(1)	Log phase	Exponential phase	Stationary phase
(2)	Lag phase	Exponential phase	Stationary phase
(3)	Exponential phase	Log phase	Stationary phase
(4)	Stationary phase	Exponential phase	Log phase

SECTION-A

Attempt All 35 questions

151. Body covered by calcareous shell and unsegmented with distinct head, muscular foot and visceral hump is seen in :-

- (1) Mollusca (2) Echinodermata
(3) Hemichordata (4) Chordata

152. Incomplete digestive system that has a single opening to outside of body that serves as both mouth and anus is present in member of which phylum ?

- (1) Porifera (2) Coelentrata
(3) Platyhelminthes (4) Aschelminthes

153. Compound epithelium is found in all **except** :

- (1) Dry surfaces of skin
(2) Moist surface of vagina
(3) Inner lining of arteries
(4) Moist surface of buccal cavity.

154. Simple squamous epithelium

- (1) Is absent in thin part of loop of Henle
(2) Is found in the walls of blood vessels and air sacs of lungs
(3) Has no role in filtration
(4) All are incorrect

155. Cuboidal epithelium with brush border of microvilli is found in :-

- (1) Eustachian tube
(2) Lining of intestine
(3) Ducts of salivary gland
(4) Proximal convoluted tubule of nephron

156. All of the following factors are favourable for dissociation of oxyhaemoglobin at tissues level, except :

- (1) High $p\text{CO}_2$
(2) Lower temperature
(3) Low $p\text{O}_2$
(4) Higher H^+ concentration

157. Mark the correct pair of muscles involved in the normal breathing in humans -

- (1) External and internal intercostal muscles
(2) Diaphragm and abdominal muscles
(3) Diaphragm and external intercostal muscles
(4) Diaphragm, external and internal intercostal muscles

158. Respiratory process is regulated by certain specialized centres in the brain. One of the following listed centres can reduce the inspiratory duration upon stimulation -

- (1) Medullary inspiratory centre
(2) Pneumotaxic centre
(3) Chemosynthetic centre
(4) Apneustic centre

159. How many Antigen is present on the surface of RBC in blood group AB positive ?

- (1) One (2) Two
(3) Three (4) Four

160. Fibrins are formed by conversion of _____ in _____ by the enzyme _____.

- (1) Fibrinogen, Serum, Thrombokinase
(2) Prothrombin, Plasma, Thrombin
(3) Fibrinogen, Plasma, Thrombin
(4) Prothrombin, blood, Thrombokinase

161. Rh antibodies from Rh -ve mother can be fatal to :-

- (1) 1st Rh $\oplus$ child
- (2) Subsequent foetus which is Rh $\ominus$ ve
- (3) Subsequent foetus which is Rh $\oplus$ ve
- (4) Not fatal to any foetus

162. Which one of the following statement is correct ?

- (1) Adult human excretes, on an average 3 to 3-5 litres of urine per day
- (2) In micturition relaxation of urethral sphincter cause release of urine
- (3) Analysis of urine have least important role in clinical diagnosis of metabolic disorders
- (4) Urine is slightly basic in nature

163. Match the column I with column II

	Column-I		Column-II
A	Nephridia	I	Prawn
B	Malpighian tubules	II	Earthworm
C	Green glands	III	Cockroach

- (1) A - I, B - II, C - III
- (2) A - III, B - II, C - I
- (3) A - II, B - III, C - I
- (4) A - II, B - I, C - III

164. Which of the following is incorrect statement about cardiac muscles?

- (1) Found in heart and intestine
- (2) Based on appearance, cardiac muscles are non-striated
- (3) They are voluntary in nature as the nervous system does not control their activities directly
- (4) All are incorrect

165. Which of the following is correct statement ?

- (1) Sternum is a flat bone on the dorsal side
- (2) The spinal cord protects the vertebrae column
- (3) Cervical vertebrae are seven in all mammals with no exception
- (4) Vertebrae column serves as the point of attachment for the musculature of the back

166. The cerebrum wraps around a structure called _____. Which is major coordinating centre for sensory and motor signaling.

- (1) Hypothalamus
- (2) Corpus callosum
- (3) Thalamus
- (4) Epithalamus

167. Find out the incorrect statement from the options given below :-

- (1) Neurotransmitter bind to their specific receptors, present on the post synaptic membrane.
- (2) A canal called the cerebral aqueduct possesses through the midbrain.
- (3) Association areas of brain are responsible for complex functions like intersensory association, memory and communication.
- (4) Electrical synapses pass electrical signal between cells with the use of Ach.

168. Select the **correct** statement w.r.t brain stem

- (1) It connects two cerebellar hemispheres
- (2) It includes midbrain and parts of hindbrain
- (3) Neural structure responsible for maintaining temperature of body
- (4) It includes midbrain and complete hindbrain

169. Which of the following statement is false ?

- (1) The endocrine system provides chemical integration through hormones.
- (2) The neural system provides an organised network of point-to-point connections for quick coordination.
- (3) In our body neural system and endocrine system jointly coordinate and integrate all the activities of the organs so that they function in a synchronised fashion.
- (4) In Hydra a brain is present along with a number of ganglia and neural tissue.

170. The posterior pituitary gland is **not** a 'true' endocrine gland because

- (1) It is provided with a duct
- (2) It only stores and releases hormones
- (3) It is under the regulation of hypothalamus
- (4) It secretes enzymes

171. Which of the following hormones can play a significant role in osteoporosis?

- (1) ADH
- (2) Aldosterone
- (3) Parathyroid hormone
- (4) Prolactin

172. **Statement I:** Oxytocin is called milk ejecting hormone.

Statement II: Epinephrine is a catecholamine.

- (1) Both statements I and II are incorrect
- (2) Statement I is correct but statement II is incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Both statements I and II are correct

173. Erythropoietin hormone which stimulates erythropoiesis is secreted by :-

- (1) β cells of pancreas
- (2) cells of Neurohypophysis
- (3) Red bone marrow
- (4) Juxta glomerular cells of kidneys

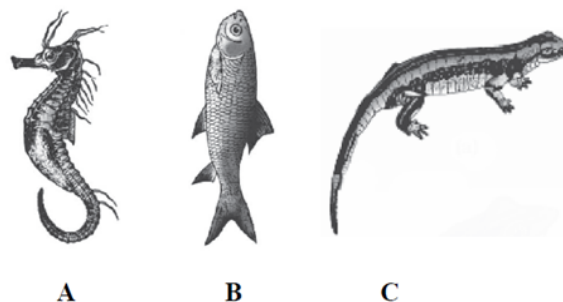
174. In amphibians, respiration occurs through -

- (1) Gills
- (2) Lungs
- (3) Skin
- (4) All of these

175. The following statement are associated with the occurrence of notochord. Identify the incorrect statement.

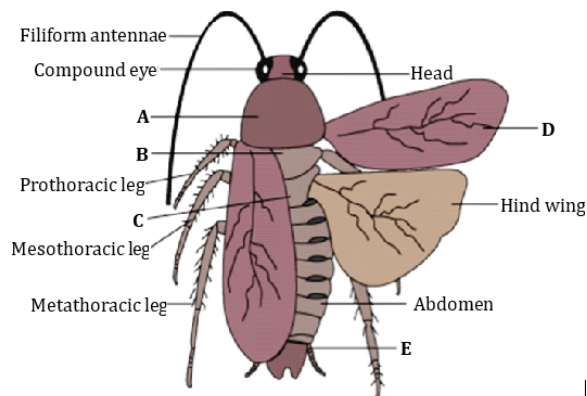
- (1) It is present only in larval tail in ascidians.
- (2) It is replaced by a vertebral column in adult frog.
- (3) It is absent throughout the life in humans from the very beginning.
- (4) It is present throughout life in Amphioxus.

176. Refer the figures A, B and C and choose the correct option which shows animals that regulate buoyancy with the help of air bladder.



- (1) A and B
- (2) A and C
- (3) B and C
- (4) All of the above

177. The following figure shows the external features of cockroach with few structures labelled as A, B, C, D, and E. Identify A to E.



- (1) A- Mesothorax, B-Pronotum, C-Metathorax, D-Tegmina, E-Anal style
- (2) A- Pronotum, B-Metathorax, C-Mesothorax-Tegmina, E-Sterna
- (3) A- Pronotum, B-Mesothorax, C-Metathorax, D-Tegmina, E-Anal cerci
- (4) A- Pronotum, B-Mesothorax, C-Metathorax, D-Tegmina, E-Anal style
178. Most oxygen is carried by the blood (i). Most carbon dioxide is carried by the blood (ii).
- (1) (i) attached to haemoglobin, (ii) in the form of bicarbonate ions.
- (2) (i) dissolved in plasma, (ii) dissolved in plasma.
- (3) (i) in the form of H^+ ions, (ii) in the form of bicarbonate ions.
- (4) (i) attached to haemoglobin, (ii) attached to haemoglobin.
179. Contraction of the ventricle in the heart begins by the command from -
- (1) Purkinje fibres
- (2) AV node
- (3) Chordae tendinae
- (4) SA node

180. During muscle contraction.

- (1) Chemical energy is changed to electrical energy.
- (2) Mechanical energy is changed to chemical energy.
- (3) Chemical energy is changed to physical energy.
- (4) Chemical energy is changed to mechanical energy.

181. 'X' is an important part of 'Y' which lies at the base of the structure which is a major coordinating centre for sensory and motor signalling. It contains a number of centre which control body temperature, urge for eating and drinking.

Identify X and Y from the options given below -

- (1) X -Cerebellum; Y – Hindbrain
- (2) X -Hypothalamus; Y - Forebrain
- (3) X -Corpora quadrigemina; Y – Midbrain
- (4) X -Pituitary gland; Y – Forebrain

182. In hormone action, if receptor molecules are removed from target organ, the target organ will -

- (1) Continue to respond to hormone.
- (2) Not respond to hormone.
- (3) Continue to respond but requires higher concentration.
- (4) Continue to respond but in the opposite way.

183. Hormones involved in carbohydrate metabolism are -

- (1) Insulin, glucagon, epinephrine and calcitonin
- (2) Insulin, glucagon, epinephrine and glucocorticoids
- (3) Insulin, glucagon, cortisol and melatonin
- (4) Insulin, glucagon, norepinephrine and melatonin

184. If Henle's loop were absent from mammalian nephron which of the following event is to be expected?

- (1) There will be no urine formation.
- (2) There will be hardly any change in the quality and quantity of urine formed.
- (3) The urine will be more concentrated.
- (4) The urine will be more dilute.

185. Which of the following statement is incorrect?

- (1) Counter-current flow of blood in vasa recta helps to retain the reabsorbed sodium in the renal medulla.
- (2) Glomerular filtrate is protein free plasma.
- (3) Vasa recta carry glomerular filtrate from distal convoluted tubule to the collecting duct.
- (4) Glomerular filtrate in Bowman's capsule is isotonic to the plasma.

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. Read the following terminology of column-I & II and identify their correct match :-

Column-I		Column-II	
(A)	Dorsal hollow nerve cord	(i)	Echinodermata
(B)	Stomochord	(ii)	Porifera
(C)	Radial symmetry	(iii)	Platyhelminthes
(D)	Feather like gills	(iv)	Annelida
		(v)	Chordata
		(vi)	Mollusca
		(vii)	Hemichordata

- (1) (A)-(v), (B)-(vii), (C)-(ii), (D)-(iv)
- (2) (A)-(v), (B)-(vii), (C)-(vi), (D)-(ii)
- (3) (A)-(v), (B)-(vii), (C)-(i), (D)-(vi)
- (4) (A)-(vii), (B)-(v), (C)-(iii), (D)-(i)

187. **Statement-1** :- The skin of Aves is unique in possessing hairs.

Statement-2 :- Most unique mammalian character is presence of mammary glands.

- (1) **Statement 1** is incorrect but **Statement 2** is correct.
- (2) Both **Statement 1** and **Statement 2** are incorrect.
- (3) Both **Statement 1** and **Statement 2** are correct.
- (4) **Statement 1** is correct but **Statement 2** is incorrect.

188. Given below are some statements :

I. Members of this class are ectoparasite on some fishes.

II. Bearing 6-15 pairs of gill slits.

III. Circular mouth without jaw.

IV. Member of this class are marine but migrate for spawning to fresh water.

How many statements belong to class cyclostomata ?

- (1) Only two
- (2) Only three
- (3) Only one
- (4) All four

189. Which of the following statements is correct for the organism shown in the given figure.



- (1) It does not shows sexual dimorphism
- (2) Body divisible into Head, Trunk and Tail
- (3) Tadpole undergoes metamorphosis to form the adult.
- (4) Female lays 20000–22000 ova at a time

190. Match the following columns

Place		pO ₂		pCO ₂	
A	Alveoli	P	95	I	40
B	Oxy. Blood	Q	40	II	45
C	Tissue	R	104	III	40

- (1) A - R - I, B - P - III, C - Q - II
- (2) A - P - II, B - Q - I, C - R - III
- (3) A - Q - I, B - P - II, C - R - III
- (4) A - R - III, B - Q - II, C - P - I

191. Find out the correct statement about human circulatory system ?

- (1) Opening between right atrium and right ventricle is guarded by tricuspid valve
- (2) Heart is ectodermally derived organ
- (3) Not all vertebrates possess muscular chambered heart
- (4) All molluscs have closed network of blood vessels

192. In which one of the following options Column I is correctly matched with Column II?

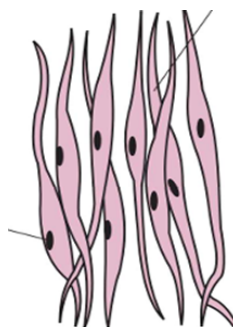
	Column I	Column II
(1)	Reabsorption of glucose, amino acids, Na ⁺	Passive
(2)	Reabsorption of urea	Active
(3)	Tubular secretion of H ⁺	Active
(4)	Conditional reabsorption of Na ⁺ and water	PCT

193. Which of the given regulatory mechanisms can be observed once there is an increase in volume of body fluid ?

- a. Osmoreceptors on hypothalamus switch off
- b. Vasopressin from neurohypophysis is released
- c. Aldosterone causes reabsorption of Na⁺ and water
- d. Constriction of blood vessels occur

- (1) a, c only
- (2) b, c, d only
- (3) a, d only
- (4) Only a

194.



Where among the following, one can find the given tissue

- (1) Stomach, intestine, heart
- (2) Intestine, stomach, biceps
- (3) Anterior 1/3rd of oesophagus, diaphragm
- (4) Stomach, uterus, urinary bladder

195. Match the list-I with list-II :-

	List-I		List-II
A.	Site of information processing and control	I.	Somatic neural system
B.	Relays impulses from the CNS to skeletal muscles	II.	Autonomic neural system
C.	Comprises of all the nerves of the body associated with the CNS	III.	Central neural system
D.	Transmits impulses from the CNS to the involuntary organs	IV.	Peripheral neural system

Choose the correct answer from the option given below :-

- (1) A-(III), B-(I), C-(IV), D-(II)
- (2) A-(III), B-(II), C-(IV), D-(I)
- (3) A-(IV), B-(I), C-(II), D-(III)
- (4) A-(I), B-(II), C-(III), D-(IV)

196. How many statement are correct regarding hormones produced by β cells of "islet of Langerhans" :-

- (a) It is a peptide hormone.
- (b) It mainly act on hepatocytes and enhance cellular glucose uptake and utilisation.
- (c) Stimulate conversion of glucose to glycogen.
- (d) Deficiency of this hormone causes diabetes mellitus.

- (1) (a), (b) is correct but (c), (d) are incorrect.
- (2) (a), (b), (c) are correct but (d) is incorrect.
- (3) (a), (b) is incorrect but (c), (d) is correct.
- (4) (a), (b), (c) and (d) are correct.

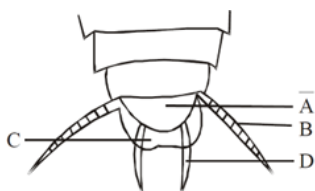
197. Read the following statements.

Statement I : Parathyroid hormone acts on bones and structure the process of bone resorption.

Statement II : Parathyroid hormone also stimulates reabsorption of Ca^{2+} by the renal tubules and increases Ca^{2+} absorption from the digested food.

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both statement I and II are correct.
- (4) Both statement I and II are incorrect.

198. The diagram given below represents the reproductive organ of male cockroach. Choose the correct labelling of the part of marked as A, B, C and D.



- (1) A – 8th sternum, B - Anal cercus, C - 10th tergum, D - Anal style
- (2) A - 10th tergum, B - Anal cercus, C - Anal style, D - 8th sternum
- (3) A - Anal style, B - Anal cercus, C - 10th tergum, D- 8th sternum
- (4) A - Anal cercus, B - 8th sternum, C - 10th tergum, D - Anal style

199. Thoracic chamber is formed (A) by the vertebral column, (B) by the sternum, (C) by the ribs and on the (D) side by the dome shaped diaphragm. Identify A, B, C and D.

- (1) A – dorsally, B – ventrally, C – laterally, D – lower
- (2) A – ventrally, B – laterally, C – dorsally, D – upper
- (3) A – laterally, B - ventrally, C – dorsally, D – lower
- (4) A – dorsally, B - laterally, C – ventrally, D – upper

200. Which one of the following statements is correct regarding blood pressure?

- (1) 190/110 mmHg may harm vital organs like brain and kidney.
- (2) 130/90 mmHg is considered high and requires treatment.
- (3) 120/80 mmHg is considered an ideal blood pressure.
- (4) 105/50 mmHg makes one very active.

